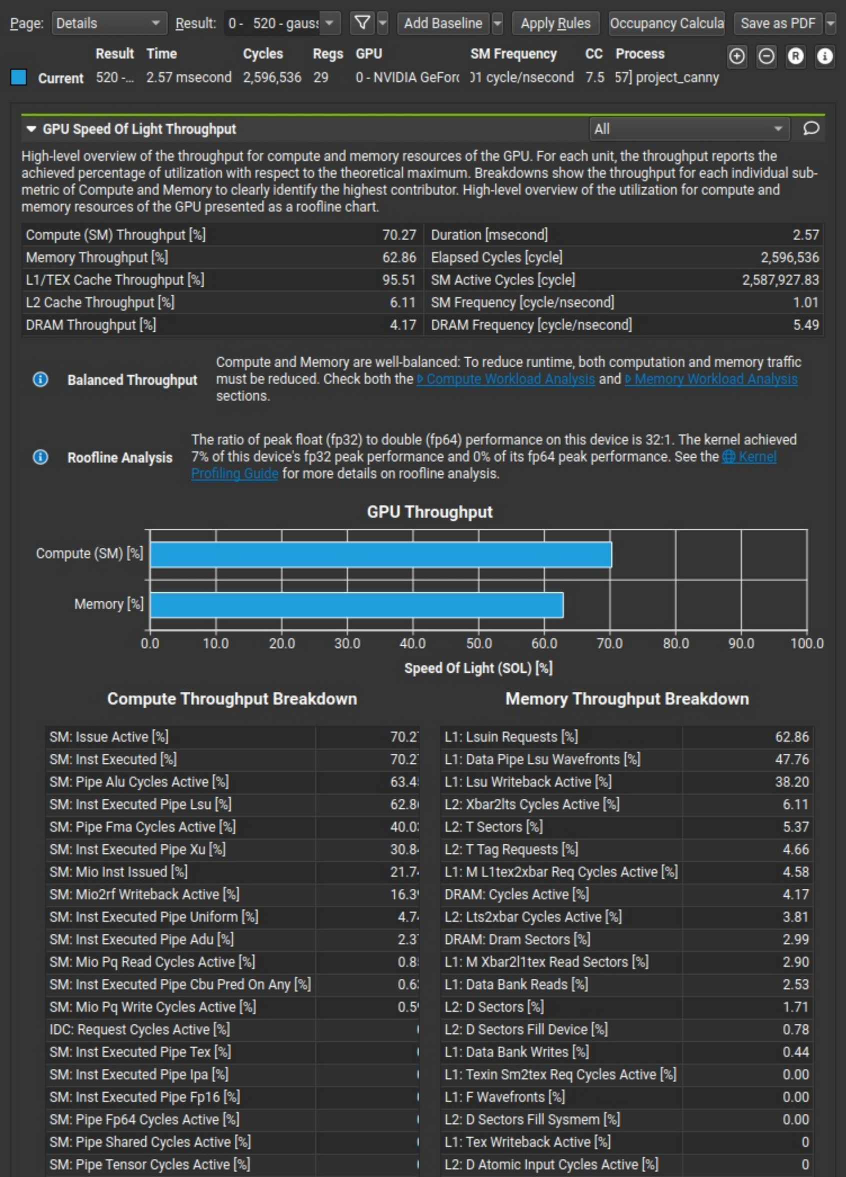

Page: Details
Result: 0 - 520 - gauss
Add Baseline
Apply Rules
Occupancy Calcula
Save as PDF
All

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
Current	520 -...	2.57 msecond	2,596,536	29	0 - NVIDIA GeFor(	)1 cycle/nsecond	7.5	57] project_canny

GPU Speed Of Light Throughput

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	70.27	Duration [msecond]	2.57
Memory Throughput [%]	62.86	Elapsed Cycles [cycle]	2,596,536
L1/TEX Cache Throughput [%]	95.51	SM Active Cycles [cycle]	2,587,927.83
L2 Cache Throughput [%]	6.11	SM Frequency [cycle/nsecond]	1.01
DRAM Throughput [%]	4.17	DRAM Frequency [cycle/nsecond]	5.49

Balanced Throughput Compute and Memory are well-balanced: To reduce runtime, both computation and memory traffic must be reduced. Check both the Compute Workload Analysis and Memory Workload Analysis sections.

Roofline Analysis The ratio of peak float (fp32) to double (fp64) performance on this device is 32:1. The kernel achieved 7% of this device's fp32 peak performance and 0% of its fp64 peak performance. See the Kernel Profiling Guide for more details on roofline analysis.

GPU Throughput

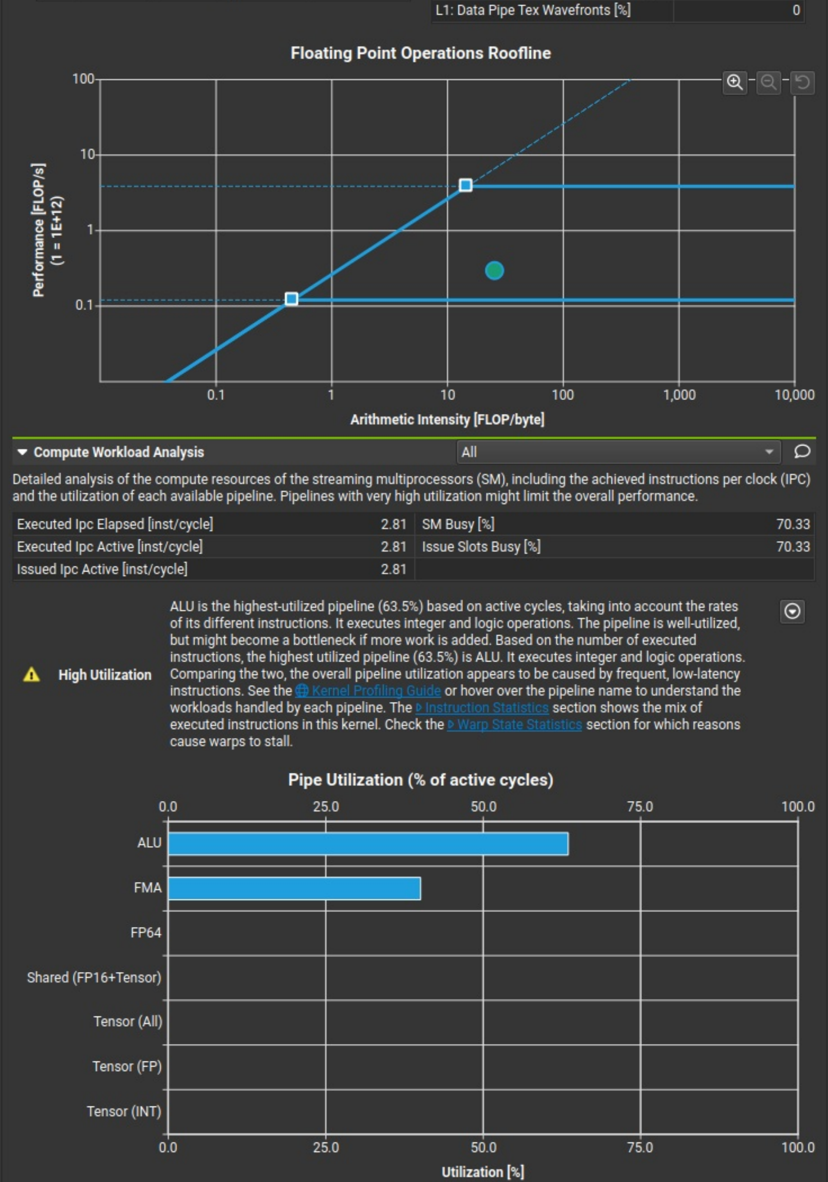
Compute (SM) [%]
Memory [%]
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
Speed Of Light (SOL) [%]

Compute Throughput Breakdown

SM: Issue Active [%]	70.2
SM: Inst Executed [%]	70.2
SM: Pipe Alu Cycles Active [%]	63.4
SM: Inst Executed Pipe Lsu [%]	62.8
SM: Pipe Fma Cycles Active [%]	40.0
SM: Inst Executed Pipe Xu [%]	30.8
SM: Mio Inst Issued [%]	21.7
SM: Mio2rf Writeback Active [%]	16.3
SM: Inst Executed Pipe Uniform [%]	4.7
SM: Inst Executed Pipe Adu [%]	2.3
SM: Mio Pq Read Cycles Active [%]	0.8
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.6
SM: Mio Pq Write Cycles Active [%]	0.5
IDC: Request Cycles Active [%]	
SM: Inst Executed Pipe Tex [%]	
SM: Inst Executed Pipe Ipa [%]	
SM: Inst Executed Pipe Fp16 [%]	
SM: Pipe Fp64 Cycles Active [%]	
SM: Pipe Shared Cycles Active [%]	
SM: Pipe Tensor Cycles Active [%]	

Memory Throughput Breakdown

L1: Lsuin Requests [%]	62.86
L1: Data Pipe Lsu Wavefronts [%]	47.76
L1: Lsu Writeback Active [%]	38.20
L2: Xbar2lts Cycles Active [%]	6.11
L2: T Sectors [%]	5.37
L2: T Tag Requests [%]	4.66
L1: M L1tex2xbar Req Cycles Active [%]	4.58
DRAM: Cycles Active [%]	4.17
L2: Lts2xbar Cycles Active [%]	3.81
DRAM: Dram Sectors [%]	2.99
L1: M Xbar2l1tex Read Sectors [%]	2.90
L1: Data Bank Reads [%]	2.53
L2: D Sectors [%]	1.71
L2: D Sectors Fill Device [%]	0.78
L1: Data Bank Writes [%]	0.44
L1: Texin Sm2tex Req Cycles Active [%]	0.00
L1: F Wavefronts [%]	0.00
L2: D Sectors Fill Sysmem [%]	0.00
L1: Tex Writeback Active [%]	0
L2: D Atomic Input Cycles Active [%]	0



L1: Data Pipe Tex Wavefronts [%]	0

Floating Point Operations Roofline

Performance [FLOP/s] (1 = 1E+12)
100
10
1
0.1
0.1
1
10
100
1,000
10,000
Arithmetic Intensity [FLOP/byte]

Compute Workload Analysis

All

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

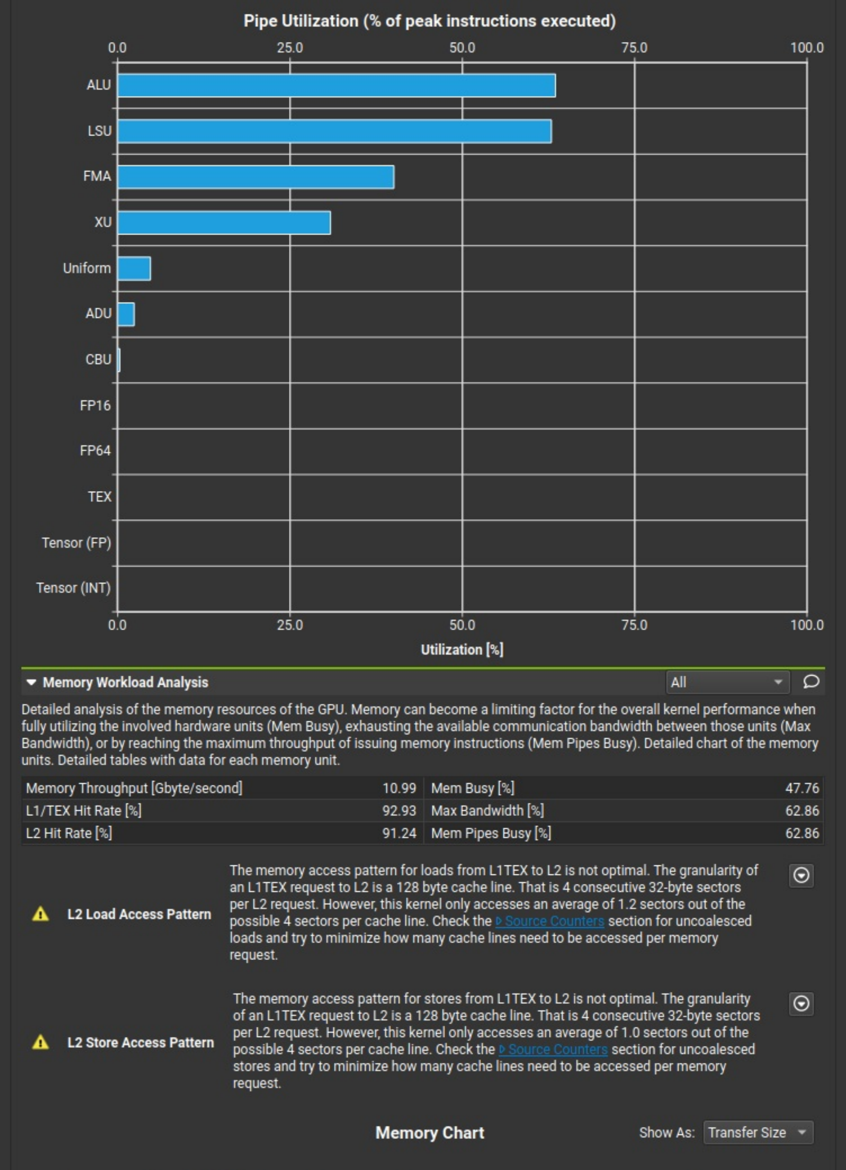
Executed Ipc Elapsed [inst/cycle]	2.81	SM Busy [%]	70.33
Executed Ipc Active [inst/cycle]	2.81	Issue Slots Busy [%]	70.33
Issued Ipc Active [inst/cycle]	2.81		

⚠ **High Utilization**

ALU is the highest-utilized pipeline (63.5%) based on active cycles, taking into account the rates of its different instructions. It executes integer and logic operations. The pipeline is well-utilized, but might become a bottleneck if more work is added. Based on the number of executed instructions, the highest utilized pipeline (63.5%) is ALU. It executes integer and logic operations. Comparing the two, the overall pipeline utilization appears to be caused by frequent, low-latency instructions. See the Kernel Profiling Guide or hover over the pipeline name to understand the workloads handled by each pipeline. The Instruction Statistics section shows the mix of executed instructions in this kernel. Check the Warp State Statistics section for which reasons cause warps to stall.

Pipe Utilization (% of active cycles)

0.0
25.0
50.0
75.0
100.0
ALU
FMA
FP64
Shared (FP16+Tensor)
Tensor (All)
Tensor (FP)
Tensor (INT)
0.0
25.0
50.0
75.0
100.0
Utilization [%]


Pipe Utilization (% of peak instructions executed)
0.0
25.0
50.0
75.0
100.0
ALU
LSU
FMA
XU
Uniform
ADU
CBU
FP16
FP64
TEX
Tensor (FP)
Tensor (INT)
Utilization [%]

Memory Workload Analysis

All

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/second]	10.99	Mem Busy [%]	47.76
L1/TEX Hit Rate [%]	92.93	Max Bandwidth [%]	62.86
L2 Hit Rate [%]	91.24	Mem Pipes Busy [%]	62.86

⚠ **L2 Load Access Pattern**

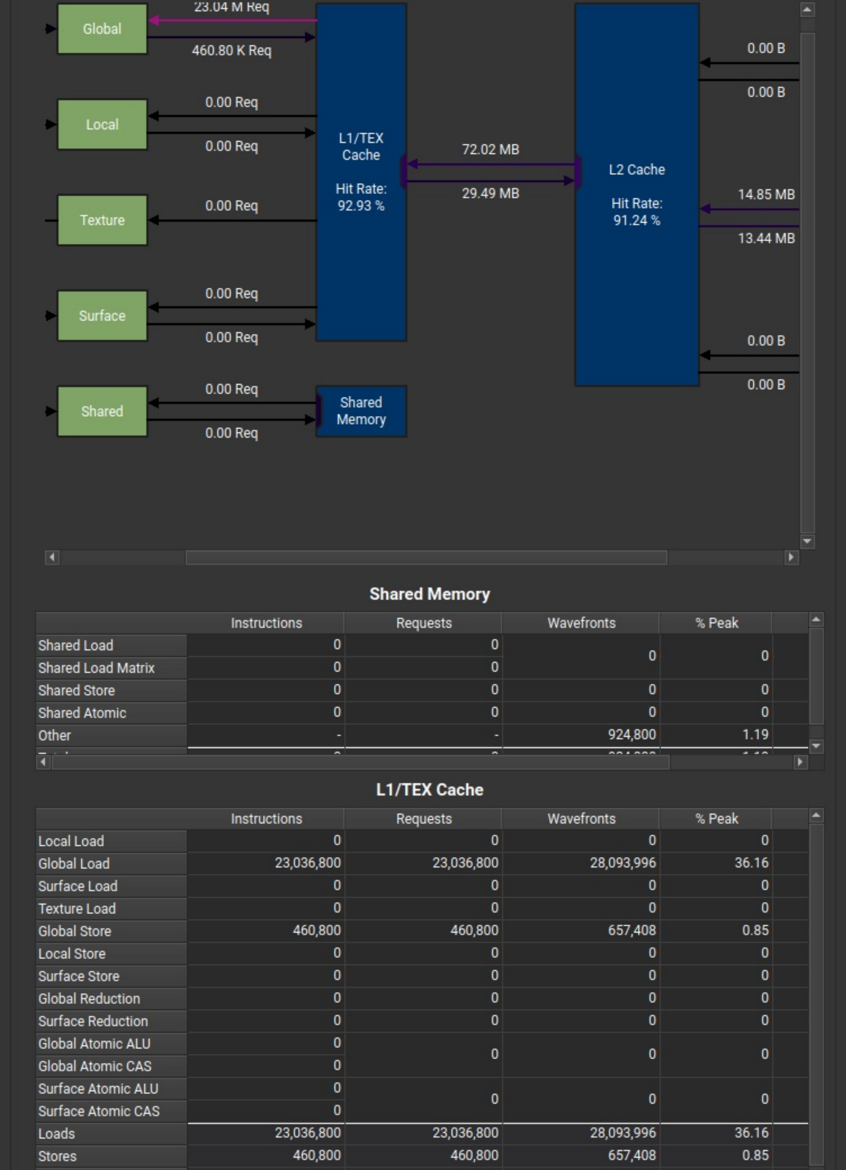
The memory access pattern for loads from L1TEX to L2 is not optimal. The granularity of an L1TEX request to L2 is a 128 byte cache line. That is 4 consecutive 32-byte sectors per L2 request. However, this kernel only accesses an average of 1.2 sectors out of the possible 4 sectors per cache line. Check the ▹ Source Counters section for uncoalesced loads and try to minimize how many cache lines need to be accessed per memory request.

⚠ **L2 Store Access Pattern**

The memory access pattern for stores from L1TEX to L2 is not optimal. The granularity of an L1TEX request to L2 is a 128 byte cache line. That is 4 consecutive 32-byte sectors per L2 request. However, this kernel only accesses an average of 1.0 sectors out of the possible 4 sectors per cache line. Check the ▹ Source Counters section for uncoalesced stores and try to minimize how many cache lines need to be accessed per memory request.

Memory Chart

Show As: Transfer Size

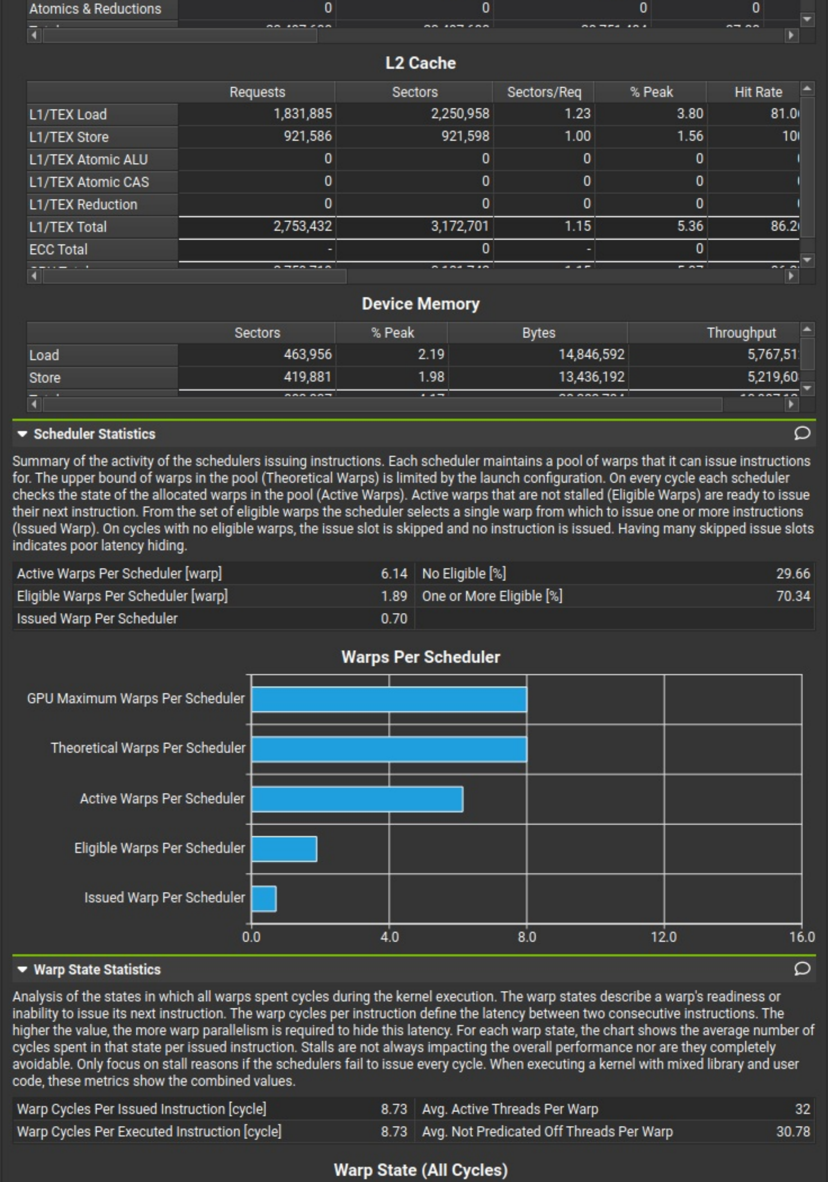

Global
23.04 M Req
460.80 K Req
Local
0.00 Req
0.00 Req
Texture
0.00 Req
Surface
0.00 Req
0.00 Req
Shared
0.00 Req
0.00 Req
L1/TEX Cache
Hit Rate: 92.93 %
Shared Memory
72.02 MB
29.49 MB
L2 Cache
Hit Rate: 91.24 %
0.00 B
0.00 B
14.85 MB
13.44 MB
0.00 B
0.00 B

Shared Memory

	Instructions	Requests	Wavefronts	% Peak
Shared Load	0	0	0	0
Shared Load Matrix	0	0		
Shared Store	0	0	0	0
Shared Atomic	0	0	0	0
Other	-	-	924,800	1.19

L1/TEX Cache

	Instructions	Requests	Wavefronts	% Peak
Local Load	0	0	0	0
Global Load	23,036,800	23,036,800	28,093,996	36.16
Surface Load	0	0	0	0
Texture Load	0	0	0	0
Global Store	460,800	460,800	657,408	0.85
Local Store	0	0	0	0
Surface Store	0	0	0	0
Global Reduction	0	0	0	0
Surface Reduction	0	0	0	0
Global Atomic ALU	0	0	0	0
Global Atomic CAS	0			
Surface Atomic ALU	0	0	0	0
Surface Atomic CAS	0			
Loads	23,036,800	23,036,800	28,093,996	36.16
Stores	460,800	460,800	657,408	0.85



Atomics & Reductions	0	0	0	0

L2 Cache

	Requests	Sectors	Sectors/Req	% Peak	Hit Rate
L1/TEX Load	1,831,885	2,250,958	1.23	3.80	81.0
L1/TEX Store	921,586	921,598	1.00	1.56	10
L1/TEX Atomic ALU	0	0	0	0	
L1/TEX Atomic CAS	0	0	0	0	
L1/TEX Reduction	0	0	0	0	
L1/TEX Total	2,753,432	3,172,701	1.15	5.36	86.2
ECC Total	-	0	-	0	

Device Memory

	Sectors	% Peak	Bytes	Throughput
Load	463,956	2.19	14,846,592	5,767,51
Store	419,881	1.98	13,436,192	5,219,60

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	6.14	No Eligible [%]	29.66
Eligible Warps Per Scheduler [warp]	1.89	One or More Eligible [%]	70.34
Issued Warp Per Scheduler	0.70		

Warps Per Scheduler

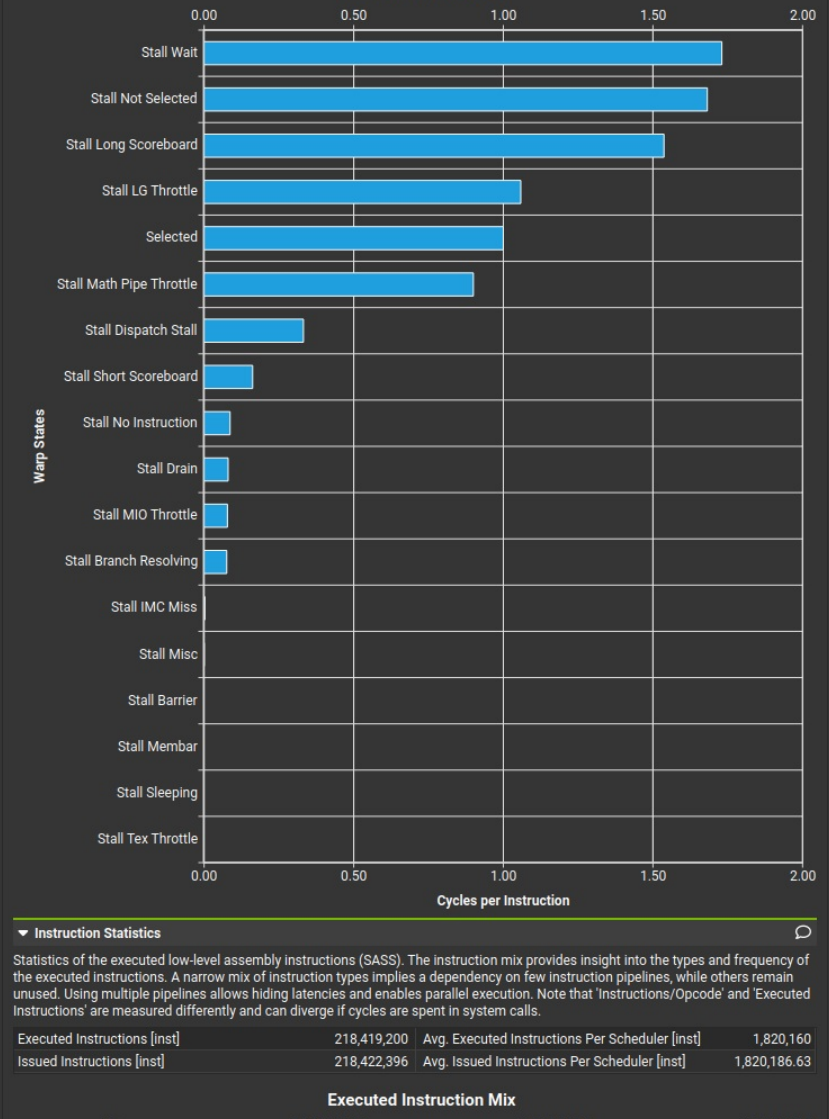
GPU Maximum Warps Per Scheduler
Theoretical Warps Per Scheduler
Active Warps Per Scheduler
Eligible Warps Per Scheduler
Issued Warp Per Scheduler
0.0
4.0
8.0
12.0
16.0

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	8.73	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	8.73	Avg. Not Predicated Off Threads Per Warp	30.78

Warp State (All Cycles)

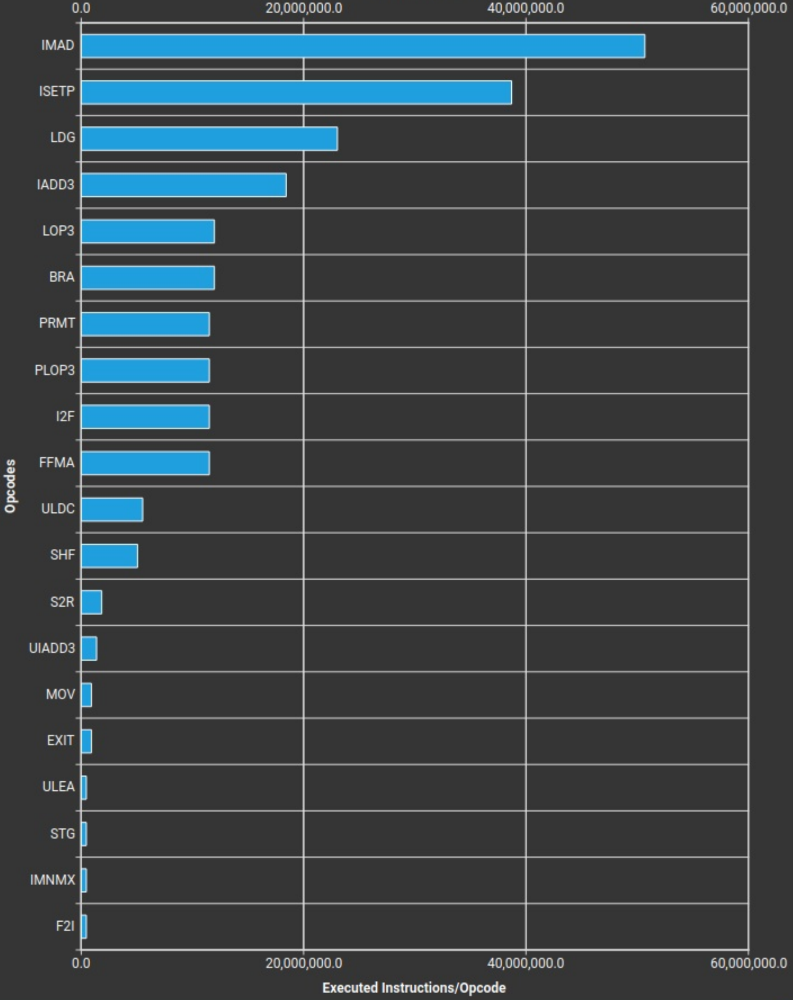

0.00
0.50
1.00
1.50
2.00
Stall Wait
Stall Not Selected
Stall Long Scoreboard
Stall LG Throttle
Selected
Stall Math Pipe Throttle
Stall Dispatch Stall
Stall Short Scoreboard
Stall No Instruction
Stall Drain
Stall MIO Throttle
Stall Branch Resolving
Stall IMC Miss
Stall Misc
Stall Barrier
Stall Membar
Stall Sleeping
Stall Tex Throttle
Warp States
Cycles per Instruction

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	218,419,200	Avg. Executed Instructions Per Scheduler [inst]	1,820,160
Issued Instructions [inst]	218,422,396	Avg. Issued Instructions Per Scheduler [inst]	1,820,186.63

Executed Instruction Mix

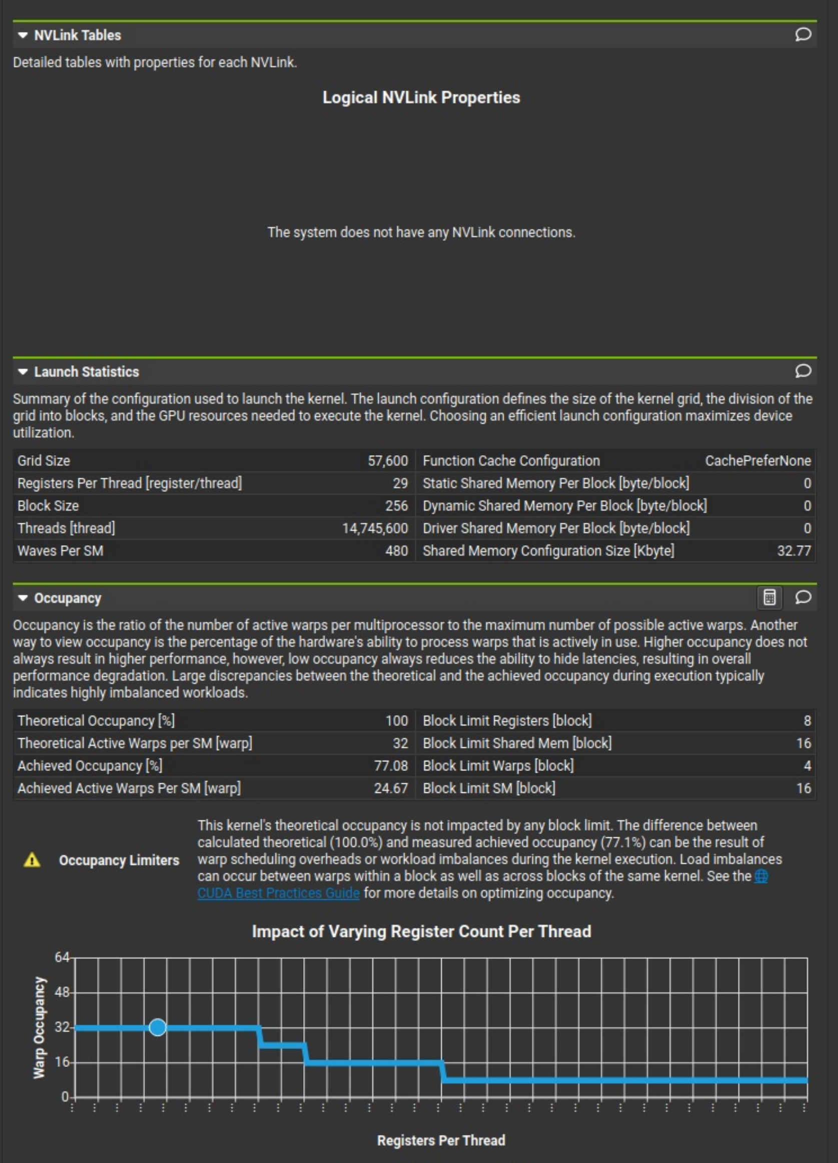


▼ NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.



NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	57,600	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	29	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	14,745,600	Driver Shared Memory Per Block [byte/block]	0
Waves Per SM	480	Shared Memory Configuration Size [Kbyte]	32.77

Occupancy

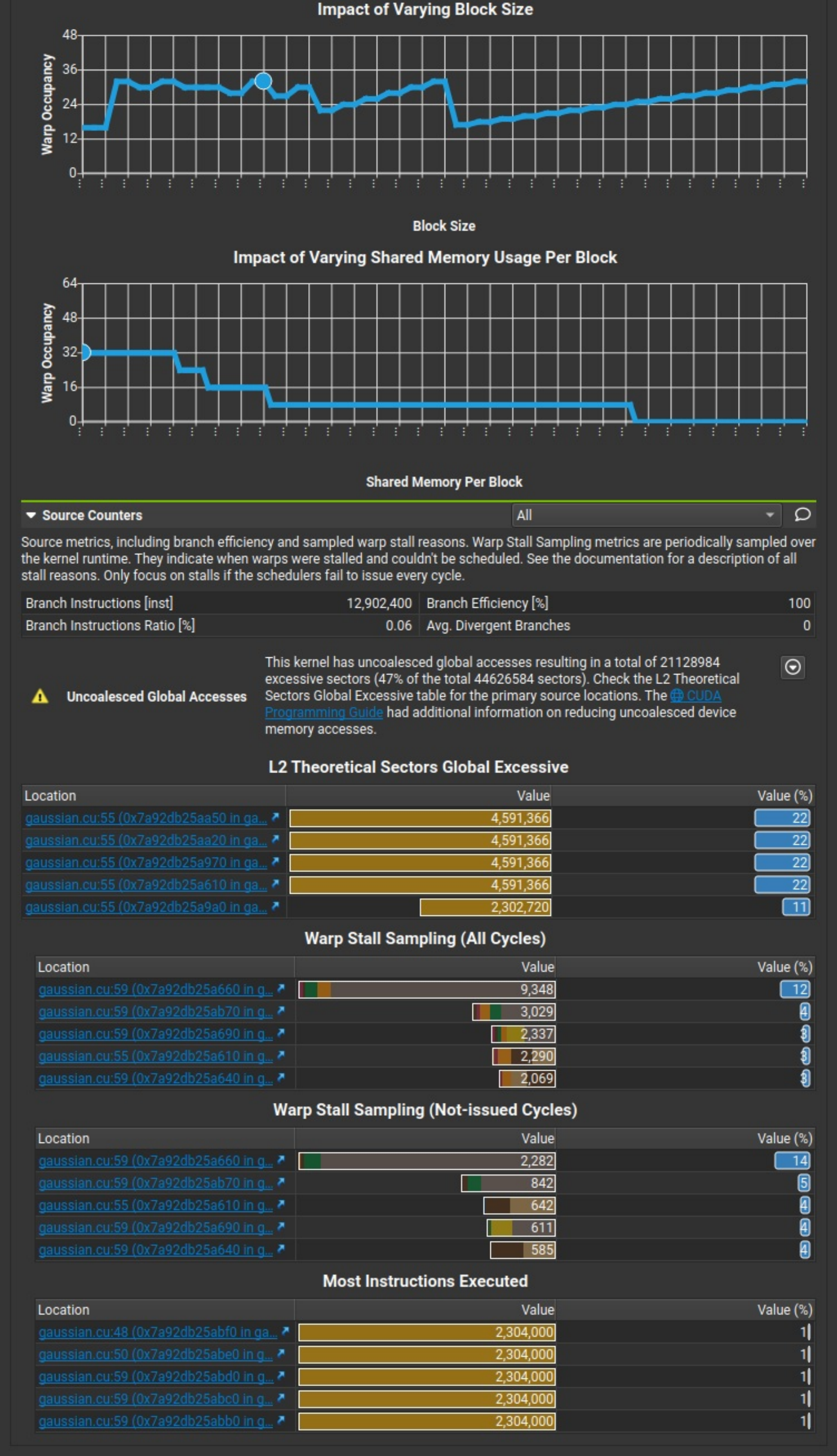
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	8
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	77.08	Block Limit Warps [block]	4
Achieved Active Warps Per SM [warp]	24.67	Block Limit SM [block]	16

⚠ **Occupancy Limiters** This kernel's theoretical occupancy is not impacted by any block limit. The difference between calculated theoretical (100.0%) and measured achieved occupancy (77.1%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the 🌐 CUDA Best Practices Guide for more details on optimizing occupancy.

Impact of Varying Register Count Per Thread

64
48
32
16
0
Warp Occupancy
Registers Per Thread


Impact of Varying Block Size
Warp Occupancy
48
36
24
12
0
Block Size
Impact of Varying Shared Memory Usage Per Block
Warp Occupancy
64
48
32
16
0
Shared Memory Per Block
Source Counters
All

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	12,902,400	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.06	Avg. Divergent Branches	0

⚠ **Uncoalesced Global Accesses**

This kernel has uncoalesced global accesses resulting in a total of 21128984 excessive sectors (47% of the total 44626584 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide had additional information on reducing uncoalesced device memory accesses.

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
gaussian.cu:55 (0x7a92db25aa50 in ga...	4,591,366	22
gaussian.cu:55 (0x7a92db25aa20 in ga...	4,591,366	22
gaussian.cu:55 (0x7a92db25a970 in ga...	4,591,366	22
gaussian.cu:55 (0x7a92db25a610 in ga...	4,591,366	22
gaussian.cu:55 (0x7a92db25a9a0 in ga...	2,302,720	11

Warp Stall Sampling (All Cycles)

Location	Value	Value (%)
gaussian.cu:59 (0x7a92db25a660 in g...	9,348	12
gaussian.cu:59 (0x7a92db25ab70 in g...	3,029	4
gaussian.cu:59 (0x7a92db25a690 in g...	2,337	3
gaussian.cu:55 (0x7a92db25a610 in g...	2,290	3
gaussian.cu:59 (0x7a92db25a640 in g...	2,069	3

Warp Stall Sampling (Not-issued Cycles)

Location	Value	Value (%)
gaussian.cu:59 (0x7a92db25a660 in g...	2,282	14
gaussian.cu:59 (0x7a92db25ab70 in g...	842	5
gaussian.cu:55 (0x7a92db25a610 in g...	642	4
gaussian.cu:59 (0x7a92db25a690 in g...	611	4
gaussian.cu:59 (0x7a92db25a640 in g...	585	4

Most Instructions Executed

Location	Value	Value (%)
gaussian.cu:48 (0x7a92db25abf0 in ga...	2,304,000	1
gaussian.cu:50 (0x7a92db25abe0 in g...	2,304,000	1
gaussian.cu:59 (0x7a92db25abd0 in g...	2,304,000	1
gaussian.cu:59 (0x7a92db25abc0 in g...	2,304,000	1
gaussian.cu:59 (0x7a92db25abb0 in g...	2,304,000	1